

Module 5: Basic Computer and Information Theory — Summary

What this module is about. Computers store representations. Binary notation uses powers of 2 instead of powers of 10, but the place-value idea is the same. Some fractions can be represented exactly in finitely many binary digits and others require approximation. Data can be compressed when it contains predictable repetition, and extra redundant bits can be added when error detection matters. Information theory measures the information in an outcome using its probability.

The binary places are

$$\dots, 2^4, 2^3, 2^2, 2^1, 2^0$$

for whole numbers and

$$2^{-1}, 2^{-2}, 2^{-3}, \dots$$

for binary fractions.

For example,

$$1101_2 = 1(8) + 1(4) + 0(2) + 1(1) = 13_{10}.$$

A byte contains 8 bits, so an unsigned byte can represent the integers 0 through 255.

Example 1: Converting a decimal number to binary

To represent 18, write it as a sum of powers of 2:

$$18 = 16 + 2 = 1(16) + 0(8) + 0(4) + 1(2) + 0(1).$$

Therefore

$$18_{10} = 10010_2.$$

The symbols changed, but the numerical value did not.

A reduced rational number has a terminating binary expansion exactly when its denominator is a power of 2. For example,

$$\frac{3}{16} = \frac{3}{2^4}$$

terminates in binary, while $3/10$ does not because the reduced denominator contains a factor of 5.

Example:

$$0.101_2 = 1\left(\frac{1}{2}\right) + 0\left(\frac{1}{4}\right) + 1\left(\frac{1}{8}\right) = \frac{5}{8} = 0.625.$$

Example 2: Error detection with even parity

Suppose the data bits are 1011. There are three 1s, which is odd. To make the total number of 1s even, append parity bit 1:

$$1011\boxed{1}.$$

If one bit flips during transmission, the receiver will see an odd number of 1s and can detect that something went wrong. Parity does not correct the message and does not detect every possible multi-bit error; it is a simple error-detection device.

Run-length encoding replaces a long repeated run by a count and a symbol. For example,

$$\text{HHHHHTTTTT} \longrightarrow 5\text{H}5\text{T}.$$

It works well when repetition is common and poorly when symbols change constantly.

The information content of an outcome with probability p is

$$I = -\log_2(p).$$

If $p = 1/32 = 2^{-5}$, then $I = 5$ bits. Each time probability is halved, information increases by 1 bit.

Module 5: Basic Computer and Information Theory — Prequiz

Open-note. Show your mathematical work. A calculation and a clearly stated final answer are enough unless a sentence is specifically requested.

Name:

Date:

Useful facts

Binary place values are powers of 2. A reduced fraction terminates in binary only when its denominator is a power of 2. Even parity makes the total number of 1s even. Information: $I = -\log_2 p$.

1. A sensor reports operating mode 13. The controller stores the mode in binary.
 - (a) Convert 13_{10} to binary.
 - (b) Convert the received binary value 10110_2 back to base ten.

2. Convert the binary fraction 0.101_2 to a base-ten decimal. Then decide whether each rational number has a finite binary expansion: $3/16$ and $3/10$. State the denominator feature you used.

3. A transmitter uses even parity.
 - (a) Data bits are 1011. What parity bit should be appended?
 - (b) A receiver gets the five-bit packet 10110. Does the parity check detect an error? Explain by counting the 1s.

4. Apply run-length encoding to the string

HHHHHTTTTT.

Then explain why the string XYZXYZXYZ is not well suited to run-length encoding.

5. An alarm event has probability $1/32$. How many bits of information does the event carry according to $I = -\log_2(p)$? Show the power-of-two step that makes the calculation immediate.

Module 5: Basic Computer and Information Theory — Matching Quiz

These problems use the same skills as the prequiz, with new values and contexts. Show enough work to make your reasoning visible.

Name:

Date:

1. Convert 22_{10} to binary, and convert 10001_2 to base ten.
2. Convert 0.011_2 to base ten. Decide whether $5/8$ and $1/5$ have finite binary expansions.
3. A transmitter uses even parity. Data bits are 0111. Determine the parity bit. Then decide whether the received packet 11011 passes the even-parity check.
4. Run-length encode MMMMMMMMMM. State whether the encoded form is shorter than writing all ten symbols.
5. One of 128 equally likely outcomes occurs. What is the probability of a particular outcome, and how many bits of information does observing it provide?